

Technology Stack

Date	29 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, JavaScript	Responsive and user-friendly interface

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask)	Processes requests and crop prediction
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MySQL	Reliable storage for crop and user data
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Google Cloud	Scalable and secure hosting
5	Version Control & CI/CD	GitHub	Version control and collaboration

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Open Weather API, Google Maps API</i>	<i>Provides real-time weather data and location-based services for accurate crop recommendations.</i>